



DEALER BEST PRACTICE SERIES

Motor Grader Ripper - Mounted Mining Bit System

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

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1.0 Introduction

Given the importance of haul road maintenance to a mine site's production, motor graders are a critical support tool to a site's operation. Some of the key tasks these machines perform include: maintaining haul roads to the original design (i.e. grade, cross-fall or crown, super-elevation, etc.), keeping rolling resistance to a minimum, and ensuring road defects such as rock spillage are removed.

To perform these tasks, the moldboard cutting edge is the primary Ground Engaging Tool (GET) utilized. However, for haul roads with harder, drier, and more compact running surfaces, such as those typically seen at sites in arid climates like Arizona and Nevada (USA), GET penetration can be more difficult when performing haul road maintenance.

2.0 Best Practice Description

This Best Practice provides an alternative use for the ripper arrangement on a motor grader. Regarding GET on a motor grader, there is typically a cutting edge bolted to the moldboard under the front frame for cutting and grading purposes. On the rear of the machine, ripper (or scarifier) tips are typically attached to the ripper shanks in the ripper arrangement to perform some lighter or less aggressive ripping tasks. However some sites, especially those located in more arid and drier climates and with harder, more compact running surfaces, may not utilize the ripper arrangement to a great degree.

For these sites and possibly others, an opportunity may exist to modify the ripper arrangement in a manner similar to that outlined below. The modification allows for a mining bit system to be installed on the ripper arrangement in place of the ripper tips.



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Motor Grader Ripper-Mounted Mining Bit System

DATE
2/16/2015

CHG
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NUMBER
1110-2.06-1262

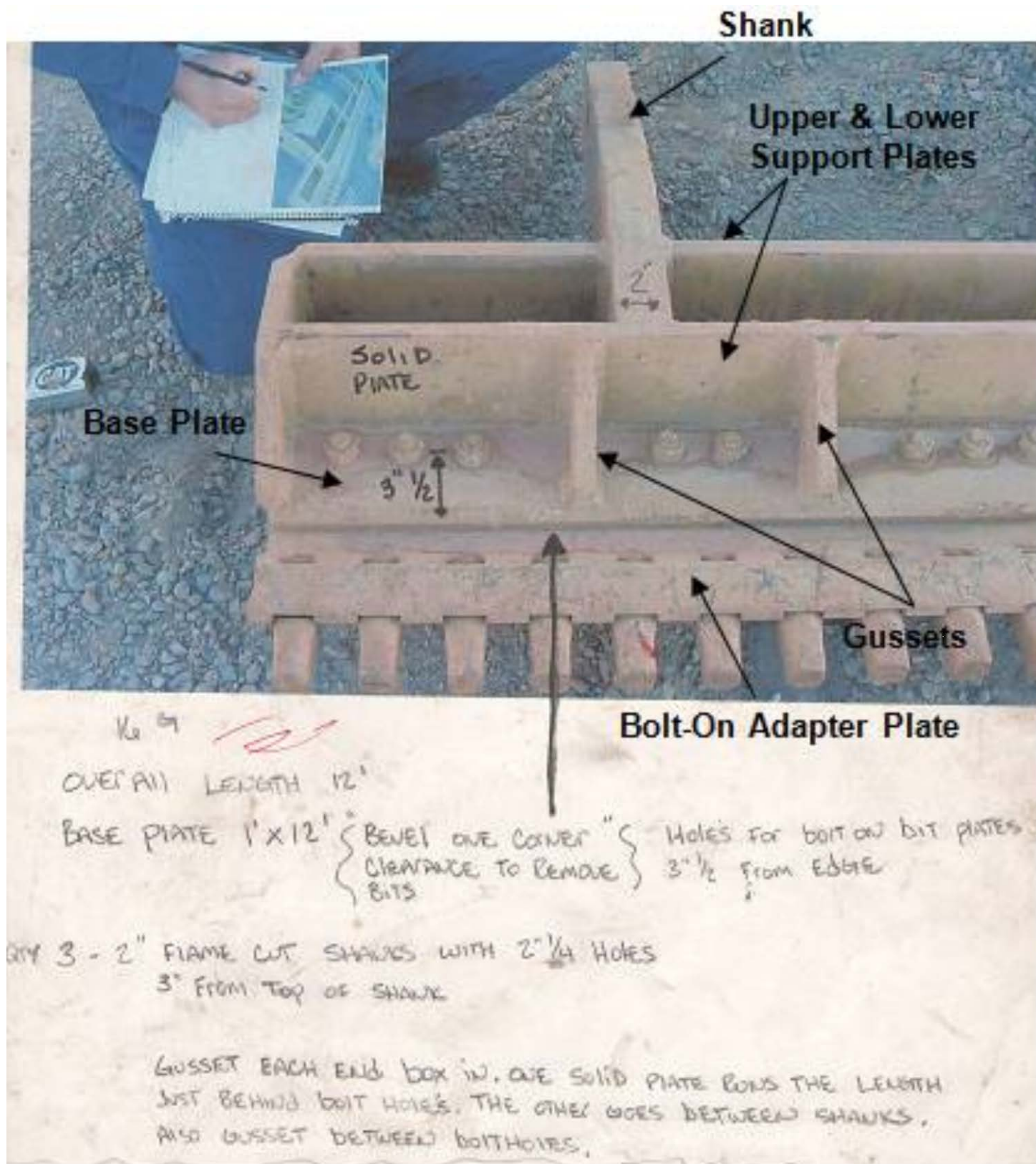
3.0 Implementation Steps

The recommended steps to follow when making this modification to the ripper arrangement are (assume 16-size grader):

- Fabricate (4) 2" flame cut shanks with 2 ¼" holes, 3" from the top of the shank. Material used for the shanks is A572 grade steel. Top corners of the shank should be rounded for easier insertion into the ripper arrangement pockets.
 - o Only (3) flame cut shanks are needed when modifying a 12-size grader.
- Fabricate the 1" by 1' by 12' base plate (also A572 grade material), with one corner beveled to facilitate bit removal. Holes for bolt-on adapter plates should be made 3 ½" from the bottom of the edge.
 - o One solid support plate (A36 grade material) runs the length of the base plate directly under the shanks and above the bolt holes; this support plate can be gusseted to the base plate to provide additional strength to the box section.
 - o Another solid support plate is welded between the shanks at the top of the base plate. The support plates and gussets are made from A36 grade material.
 - o A 20' piece of 1" by 6" flat bar is typically used for the support gussets; a 20' piece of 1" by 8" is used for the support plates and to also cap the ends of the box section.
- Weld the shanks to the base plate and upper and lower support plates, as well as weld in the gussets (if used) to provide additional support to the box section. The weld wire used is flux-cored gas-shielded 71M wire (.052" (1.3mm) diameter).
 - o Reference the following pictures for further clarity.

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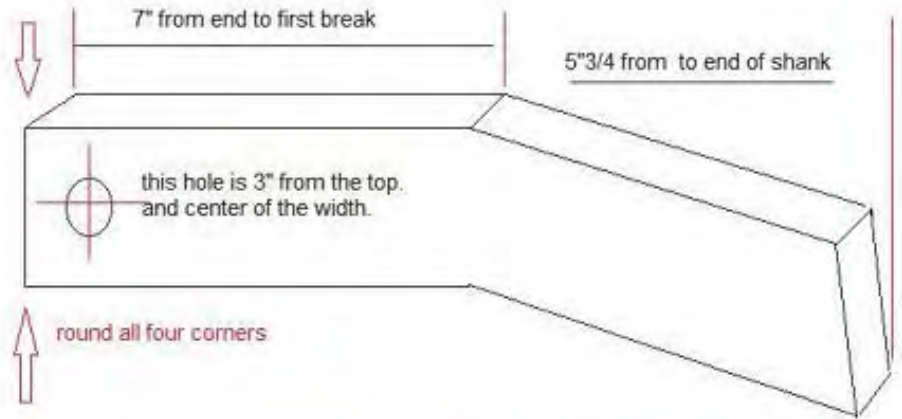
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note the shanks are 2" thick , only the lenght has been modified to a total of 12³/₄ .
the shanks are the same width as original shanks that come on the machine.
the material use to make base plate and shanks are **A572** grade material
also note that the top corners need to be rounded to better fit pockets.

the gussets and support plates are all A36 material. only the base plate and shanks are **A572**

It is important to remember the following when using the rotating bit system:

- Bit penetration should not exceed the carbide; therefore bit depth should be approximately 1.0" to 2.0" (25mm to 51mm).
- Operators should keep the bits perpendicular to the ground surface, with a cast angle between 0° and 10°.
- Operating speed should be between 4 to 6 mph (not exceeding 6 mph).



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4.0 Benefits

The benefits of this modification are the following:

- Increased effectiveness of grading capability for hard-to-penetrate haul roads, as the road surface is turned over and can then be easily graded.
- Cutting edge remains attached to machine for grading, road maintenance, and ditching work, saving removal & install time of changing out different GET systems on the grader.
- Reduction in number of cutting edges consumed as the rotating bit system is used for breaking up the hardened road surface before finishing with the cutting edge.
- Longer-wearing carbide on bits withstands severe haul road conditions.
- Easy and fast change-out of mining bits when replacement interval occurs.
- Narrow and wide gaps on mining bit system provide more flexibility on road aggregate sizing.



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5.0 Resources Required

- Access to steel flat stock for shanks, base plates, support plates, and gusset fabrication.
- Fabrication and weld shop to cut shanks and plates to length and weld base plate box section to shanks.
- Parts ordering system for rotating bit system. The GET Parts Sales Kit (PSK), 10th Edition (PECP3001 - <https://dealer.cat.com/cda/components/fullArticle?m=20031&x=7&id=3048901>) can also be referenced for part number information. Section 10 – Motor Graders contains the applicable information regarding Cat rotating bit systems.
- Assembly of rotating bit system on ripper arrangement.

6.0 Supporting Attachments / References

- GET Parts Sales Kit (10th Edition) - Section 10 for additional information regarding the mining bit system. PECJ0025-03.pdf

7.0 Related Best Practices

None.

8.0 Acknowledgements

The Caterpillar dealer Empire Southwest is specially acknowledged for contributing information related to this Best Practice. Caterpillar Global Mining extends its appreciation to Empire (specifically Andy Brown, Tom Haugen, George McKeon, and Steve Soto) for sharing this idea with the extended Cat mining team.

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